

Extended Bioprocess Model: Mixing, Growth, and Foam Dynamics

Physical & Mixing Dynamics

1. Volumetric Mass Transfer Coefficient ($k_L a$)

Sets the physical limit for oxygen transfer based on agitation power (P/V) and superficial gas velocity (v_s).

$$k_L a = K \left(\frac{P}{V} \right)^\alpha (v_s)^\beta$$

2. Mixing Time (t_m)

Evaluates vessel blending efficiency based on circulation number (N_c) and impeller speed (N).

$$t_m = \frac{N_c}{N}$$

Biological Growth & Population Kinetics

3. Oxygen-Limited Specific Growth Rate (μ_1)

Dual-Monod kinetics linking primary culture growth to both limiting substrate (S) and dissolved oxygen (C_L).

$$\mu_1 = \mu_{max,1} \left(\frac{S}{K_{s,1} + S} \right) \left(\frac{C_L}{K_{o,1} + C_L} \right)$$

4. Primary Culture Biomass Rate $\left(\frac{dX_1}{dt} \right)$

$$\frac{dX_1}{dt} = \mu_1 X_1 - k_{d1} X_1$$

5. Contaminant Biomass Rate $\left(\frac{dX_2}{dt} \right)$

$$\frac{dX_2}{dt} = \mu_2 X_2 - k_{d2} X_2$$

Substrate Consumption

6. Competitive Substrate Depletion $\left(\frac{dS}{dt}\right)$

Tracks shared resource consumption by both primary culture and contaminant populations.

$$\frac{dS}{dt} = -\frac{\mu_1 X_1}{Y_{X_1/S}} - \frac{\mu_2 X_2}{Y_{X_2/S}} - m_1 X_1 - m_2 X_2$$

Respiration & Gas Exchange

7. Oxygen Uptake Rate (*OUR*)

Total biological oxygen demand.

$$OUR = q_{O_2,1} X_1 + q_{O_2,2} X_2$$

8. Carbon Dioxide Evolution Rate (*CER*)

Total biological CO_2 production.

$$CER = q_{CO_2,1} X_1 + q_{CO_2,2} X_2$$

9. Respiration Quotient (*RQ*)

Ratio of CO_2 produced to O_2 consumed.

$$RQ = \frac{CER}{OUR}$$

10. Dissolved Oxygen Derivative $\left(\frac{dC_L}{dt}\right)$

The continuous balance between physical transfer and biological uptake.

$$\frac{dC_L}{dt} = k_L a (C^* - C_L) - OUR$$

Foam Generation & Drainage

11. Biosurfactant Concentration (C_{surf})

Tracks foam-stabilizing proteins produced by total active biomass.

$$C_{surf} = k_{yield} (X_1 + X_2)$$

12. Dynamic Foam Height $\left(\frac{dh_f}{dt}\right)$

Balances aeration-driven foam creation against natural drainage.

$$\frac{dh_f}{dt} = k_f v_s C_{surf} - k_{drain} h_f$$

13. Steady-State Foam Volume (V_f)

Based on the Bikerman equation using retention time (Σ) and gas flow rate (Q_g).

$$V_f = \Sigma Q_g$$

AI Predictors & Recoverability Engine

14. Cumulative Kinetic Stress (λ)

Integrates deviations from optimal pH and critical dissolved oxygen limits.

$$\lambda = k_{stress} |pH_{opt} - pH(t)| + k_{O_2} |C_{crit} - C_L|$$

15. Dynamic Recovery Probability ($R(t)$)

Calculates the real-time percentage (0–100%) defining whether the batch remains biologically recoverable.

$$R(t) = \exp\left(-\int \lambda dt\right)$$

Symbol reference: $k_L a$ = volumetric oxygen mass transfer coefficient; K , α , β = empirical correlation constants; P/V = agitation power per unit volume; v_s = superficial gas velocity; t_m = mixing time; N_c = circulation number; N = impeller speed; μ_1 , μ_2 = specific growth rates of primary culture and contaminant; $\mu_{max,1}$ = maximum specific growth rate of primary culture; S = substrate concentration; $K_{s,1}$ = substrate half-saturation constant; C_L = dissolved oxygen concentration; $K_{o,1}$ = oxygen half-saturation constant; X_1 , X_2 = biomass concentrations of primary culture and contaminant; k_{d1} , k_{d2} = death/decay rate constants; $Y_{X_1/S}$, $Y_{X_2/S}$ = yield coefficients on substrate; m_1 , m_2 = maintenance coefficients; $q_{O_2,1}$, $q_{O_2,2}$ = specific oxygen uptake rates; $q_{CO_2,1}$, $q_{CO_2,2}$ = specific carbon dioxide evolution rates; C^* = saturation dissolved oxygen concentration; C_{surf} = biosurfactant concentration; k_{yield} = surfactant yield constant; h_f = foam height; k_f = foam formation constant; k_{drain} = foam drainage constant; V_f = steady-state foam volume; Σ = foam retention time; Q_g = gas flow rate; λ = cumulative kinetic stress; k_{stress} , k_{O_2} = stress and oxygen-deviation sensitivity constants; pH_{opt} = optimal pH; C_{crit} = critical dissolved oxygen concentration; $R(t)$ = dynamic recovery probability.